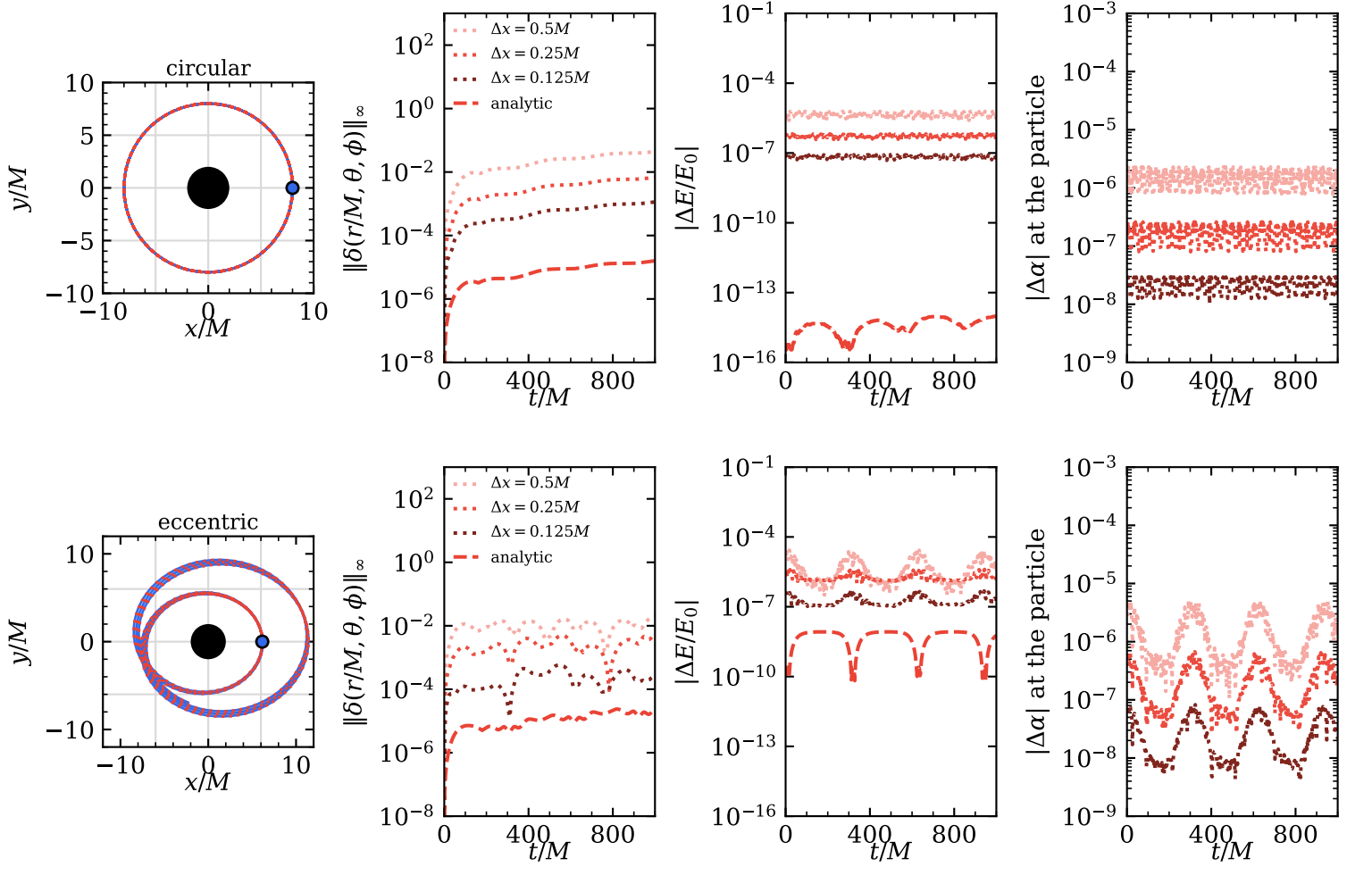

circular
eccentric
y/M
x/M
$\|\delta(r/M,\theta,\phi)\|_\infty$
$\Delta x = 0.5M$
$\Delta x = 0.25M$
$\Delta x = 0.125M$
analytic
t/M
$|\Delta E/E_0|$
$|\Delta\alpha|$ at the particle